N-Gram Language Models & Neural Networks

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https://laramartin.net/NLP-class/

Slides modified from Dr. Frank Ferraro

Learning Objectives

Create a LM using smoothed counts

Define the basic architecture of a neural network

Distinguish between count-based, logistic regression, and neural LMs

Review: Probability Chain Rule

$$p(x_1, x_2, \dots, x_S) =$$
$$p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_1, x_2) \cdots p(x_S \mid x_1, \dots, x_{S-1}) =$$
$$\prod_{i}^{S} \underbrace{p(x_i \mid x_1, \dots, x_{i-1})}$$

Language modeling is about how to estimate each of these factors in {great, good, sufficient, …} ways

Review: MLE with Trigrams

p(Colorless green ideas sleep furiously) =
p(Colorless | *<BOS> <BOS>*) *
p(green | *<BOS>* Colorless) *
p(ideas | Colorless green) *
p(sleep | green ideas) *
p(furiously | ideas sleep) *
p(*<EOS>* | sleep furiously)

Consistent notation: Pad the left with <BOS> (beginning of sentence) symbols
Fully proper distribution: Pad the right with a single <EOS> symbol

Review: Count-Based N-Grams (Unigrams)

word type

word type

$$p(\mathrm{z}) \propto count(\mathrm{z})$$

$$= \frac{count(\mathrm{z})}{\sum_{v} count(\mathrm{v})}$$

word type

Review: Count-Based N-Grams (Trigrams)

$$p(\mathrm{z}|\mathrm{x},\mathrm{y}) \propto count(\mathrm{x},\mathrm{y},\mathrm{z})$$

$$= \frac{count(\mathrm{x},\mathrm{y},\mathrm{z})}{\sum_{v} count(\mathrm{x},\mathrm{y},\mathrm{v})}$$

Types of Early LMs

Maximum likelihood (MLE): simple counting

Other count-based models

- **Laplace smoothing, add- λ**
- Interpolation models
- Discounted backoff
- Interpolated (modified) Kneser-Ney
- Good-Turing
- Witten-Bell

Easy to implement

Advanced/ out of scope

Maxent n-gram models

Featureful LMs

Neural n-gram models

Feedforward LMs

Recurrent/autoregressive NNs

Precursor to modern LMs

Add-λ estimation

Other names: Laplace smoothing, Lidstone smoothing

Pretend we saw each word λ more times than we did

$$p(\mathrm{z}) \propto count(\mathrm{z}) + \lambda$$

Add λ to all the counts

Add-λ estimation

Other names: Laplace smoothing, Lidstone smoothing

Pretend we saw each word λ more times than we did

Add λ to all the counts

$$p(\mathrm{z}) \propto count(\mathrm{z}) + \lambda$$

$$= \frac{count(\mathrm{z}) + \lambda}{\sum_v (count(\mathrm{v}) + \lambda)}$$

Add-λ estimation

Other names: Laplace smoothing, Lidstone smoothing

Pretend we saw each word λ more times than we did

Add λ to all the counts

$$p(\mathrm{z}) \propto count(\mathrm{z}) + \lambda$$

$$= \frac{count(\mathrm{z}) + \lambda}{W + V\lambda}$$

tokens

types

Add-λ N-Grams (Unigrams)

The film got a great opening and the film went on to become a hit .

Word (Type)	Raw Count	Norm	Prob.	Add-λ Count	Add-λ Norm.	Add-λ Prob.
The	1	16	1/16			
film	2		1/8			
got	1		1/16			
a	2		1/8			
great	1		1/16			
opening	1		1/16			
and	1		1/16			
the	1		1/16			
went	1		1/16			
on	1		1/16			
to	1		1/16			
become	1		1/16			
hit	1		1/16			
.	1		1/16			

Add-1 N-Grams (Unigrams)

The film got a great opening and the film went on to become a hit .

Word (Type)	Raw Count	Norm	Prob.	Add-1 Count	Add-1 Norm.	Add-1 Prob.
The	1	16	1/16	2		
film	2		1/8	3		
got	1		1/16	2		
a	2		1/8	3		
great	1		1/16	2		
opening	1		1/16	2		
and	1		1/16	2		
the	1		1/16	2		
went	1		1/16	2		
on	1		1/16	2		
to	1		1/16	2		
become	1		1/16	2		
hit	1		1/16	2		
.	1		1/16	2		

Add-1 N-Grams (Unigrams)

The film got a great opening and the film went on to become a hit .

Word (Type)	Raw Count	Norm	Prob.	Add-1 Count	Add-1 Norm.	Add-1 Prob.
The	1	16	1/16	2	16 + 14*1 = 30	
film	2		1/8	3		
got	1		1/16	2		
a	2		1/8	3		
great	1		1/16	2		
opening	1		1/16	2		
and	1		1/16	2		
the	1		1/16	2		
went	1		1/16	2		
on	1		1/16	2		
to	1		1/16	2		
become	1		1/16	2		
hit	1		1/16	2		
.	1		1/16	2		

Add-1 N-Grams (Unigrams)

The film got a great opening and the film went on to become a hit .

Word (Type)	Raw Count	Norm	Prob.	Add-1 Count	Add-1 Norm.	Add-1 Prob.
The	1	16	1/16	2	16 + 14*1 = 30	=1/15
film	2		1/8	3		=1/10
got	1		1/16	2		=1/15
a	2		1/8	3		=1/10
great	1		1/16	2		=1/15
opening	1		1/16	2		=1/15
and	1		1/16	2		=1/15
the	1		1/16	2		=1/15
went	1		1/16	2		=1/15
on	1		1/16	2		=1/15
to	1		1/16	2		=1/15
become	1		1/16	2		=1/15
hit	1		1/16	2		=1/15
.	1		1/16	2		=1/15

An Extended Trigram Example

The film got a great opening and the film went on to become a hit .

Q: With OOV, EOS, and BOS, how many types (for normalization)?

Context: x y	Word (Type): z	Raw Count	Add-1 count	Norm.	Probability p(z | x y)
The film	The	0			
The film	film	0			
The film	got	1			
The film	went	0			
The film	OOV	0			
The film	EOS	0			
…					
a great	great	0			
a great	opening	1			
a great	and	0			
a great	the	0			
…					

An Extended Trigram Example

The film got a great opening and the film went on to become a hit .

Q: With OOV, EOS, and BOS, how many types (for normalization)?

A: 16
(why don't we count BOS?)

Context: x y	Word (Type): z	Raw Count	Add-1 count	Norm.	Probability p(z | x y)
The film	The	0			
The film	film	0			
The film	got	1			
The film	went	0			
The film	OOV	0			
The film	EOS	0			
…					
a great	great	0			
a great	opening	1			
a great	and	0			
a great	the	0			
…					

An Extended Trigram Example

The film got a great opening and the film went on to become a hit .

Q: With OOV, EOS, and BOS, how many types (for normalization)?

A: 16
(why don't we count BOS?)

Context: x y	Word (Type): z	Raw Count	Add-1 count	Norm.	Probability p(z | x y)
The film	The	0	1	17 (=1+16*1)	1/17
The film	film	0	1		1/17
The film	got	1	2		2/17
The film	went	0	1		1/17
…					…
The film	OOV	0	1		1/17
The film	EOS	0	1		1/17
…					
a great	great	0	1	17	1/17
a great	opening	1	2		2/17
a great	and	0	1		1/17
a great	the	0	1		1/17
…					

An Extended Trigram Example

The film got a great opening and the film went on to become a hit .

Context: x y	Word (Type): z	Raw Count	Add-1 count	Norm.	Probability p(z | x y)
The film	The	0	1	17 (=1+16*1)	1/17
The film	film	0	1		1/17
The film	got	1	2		2/17
The film	went	0	1		1/17
…					…
The film	OOV	0	1		1/17
The film	EOS	0	1		1/17
…					
a great	great	0	1	17	1/17
a great	opening	1	2		2/17
a great	and	0	1		1/17
a great	the	0	1		1/17
…					

Q: What is the perplexity for the sentence
"The film , a hit !"

What are the tri-grams for “The film , a hit !”

Trigrams	MLE p(trigram)
<BOS> <BOS> The	1
<BOS> The film	1
The film ,	0
film , a	0
, a hit	0
a hit !	0
hit ! <EOS>	0

What are the tri-grams for "The film , a hit !"

Trigrams	MLE p(trigram)	UNK-ed trigrams
<BOS> <BOS> The	1	<BOS> <BOS> The
<BOS> The film	1	<BOS> The film
The film ,	0	The film <UNK>
film , a	0	film <UNK> a
, a hit	0	<UNK> a hit
a hit !	0	a hit <UNK>
hit ! <EOS>	0	hit <UNK> <EOS>

What are the tri-grams for "The film , a hit !"

Trigrams	MLE p(trigram)	UNK-ed trigrams	Smoothed p(trigram)
<BOS> <BOS> The	1	<BOS> <BOS> The	2/17
<BOS> The film	1	<BOS> The film	2/17
The film ,	0	The film <UNK>	1/17
film , a	0	film <UNK> a	1/16
, a hit	0	<UNK> a hit	1/16
a hit !	0	a hit <UNK>	1/17
hit ! <EOS>	0	hit <UNK> <EOS>	1/16

Types of Early LMs

Maximum likelihood (MLE): simple counting

Other count-based models

- Laplace smoothing, add- λ
- Interpolation models
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- Good-Turing
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Maxent n-gram models ← Featureful LMs

Neural n-gram models ← Feedforward LMs

Recurrent/autoregressive NNs ← Precursor to modern LMs

Maxent Models as Featureful n-gram Language Models

p(Colorless green ideas sleep furiously | Label) =
p(Colorless | Label, <BOS>) * … * p(<EOS> | Label , furiously)

Model each n-gram term with a maxent model

$$p(x_i \mid y, x_{i-N+1:i-1}) = \text{maxent}(y, x_{i-N+1:i-1}, x_i)$$

generatively trained:
learn to model *(class-specific) language*

Language Model with Maxent n-grams

label

$$p_n(\text{🗎}|y) = \prod_{i=1}^{M} \text{maxent}(y, x_{i-n+1:i-1}, x_i)$$

n-gram

$$= \prod_{i=1}^{M} \frac{\exp(\theta_{x_i}^T f(y, x_{i-n+1:i-1}))}{\sum_{x'} \exp(\theta_{x'}^T f(y, x_{i-n+1:i-1}))}$$

Iterate through all possible output vocab types x'---just like in count-based LMs

What Should These Features Do?

$p(x_i \mid y, x_{i-N+1:i-1}) = \text{maxent}(y, x_{i-N+1:i-1}, x_i)$, e.g.,

$$p(\text{sleep} \mid y, \text{green}, \text{ideas}) =$$
$$\text{maxent}\big(y, x_{i-2,i-1} = (\text{green}, \text{ideas}), x_i = \text{sleep}\big)$$
$$\propto \exp(\theta_{x_i=\text{sleep}}{}^T f(y, x_{i-2,i-1} = (\text{green}, \text{ideas})))$$

(in-class discussion)

N-gram Language Models

given some context…

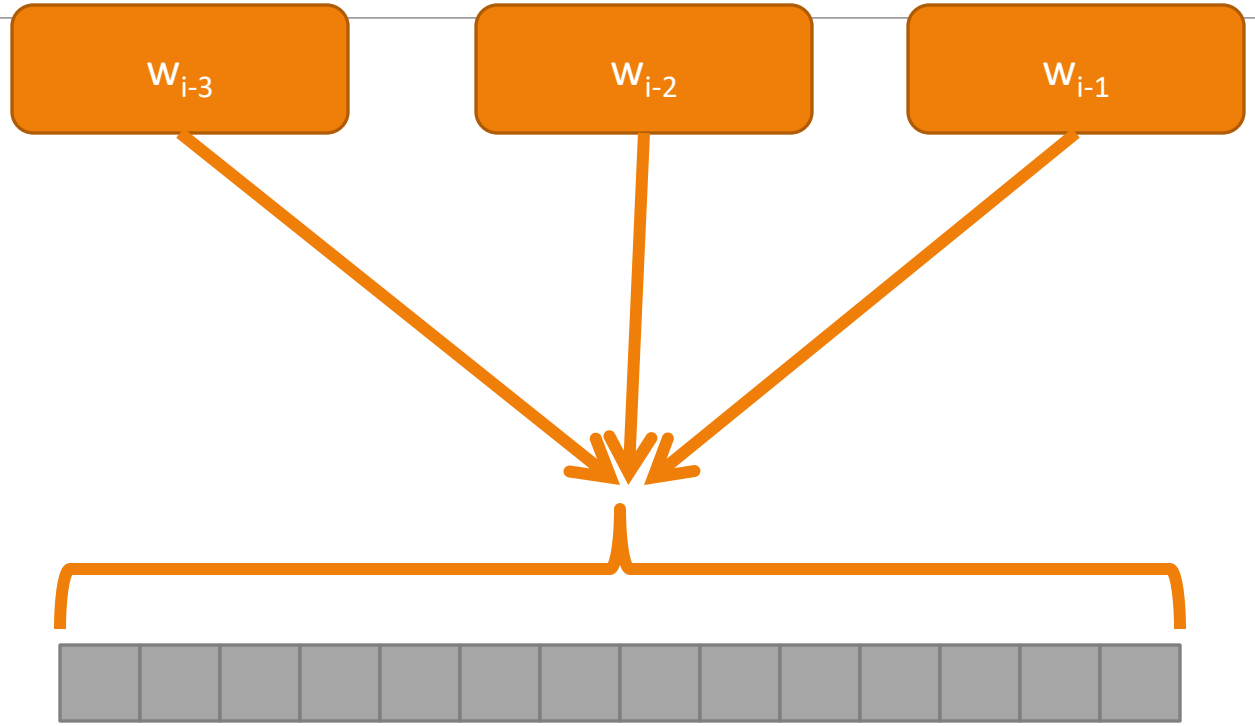
w_{i-3} w_{i-2} w_{i-1}

predict the next word

w_i

N-gram Language Models

given some context…



compute beliefs about what is likely…

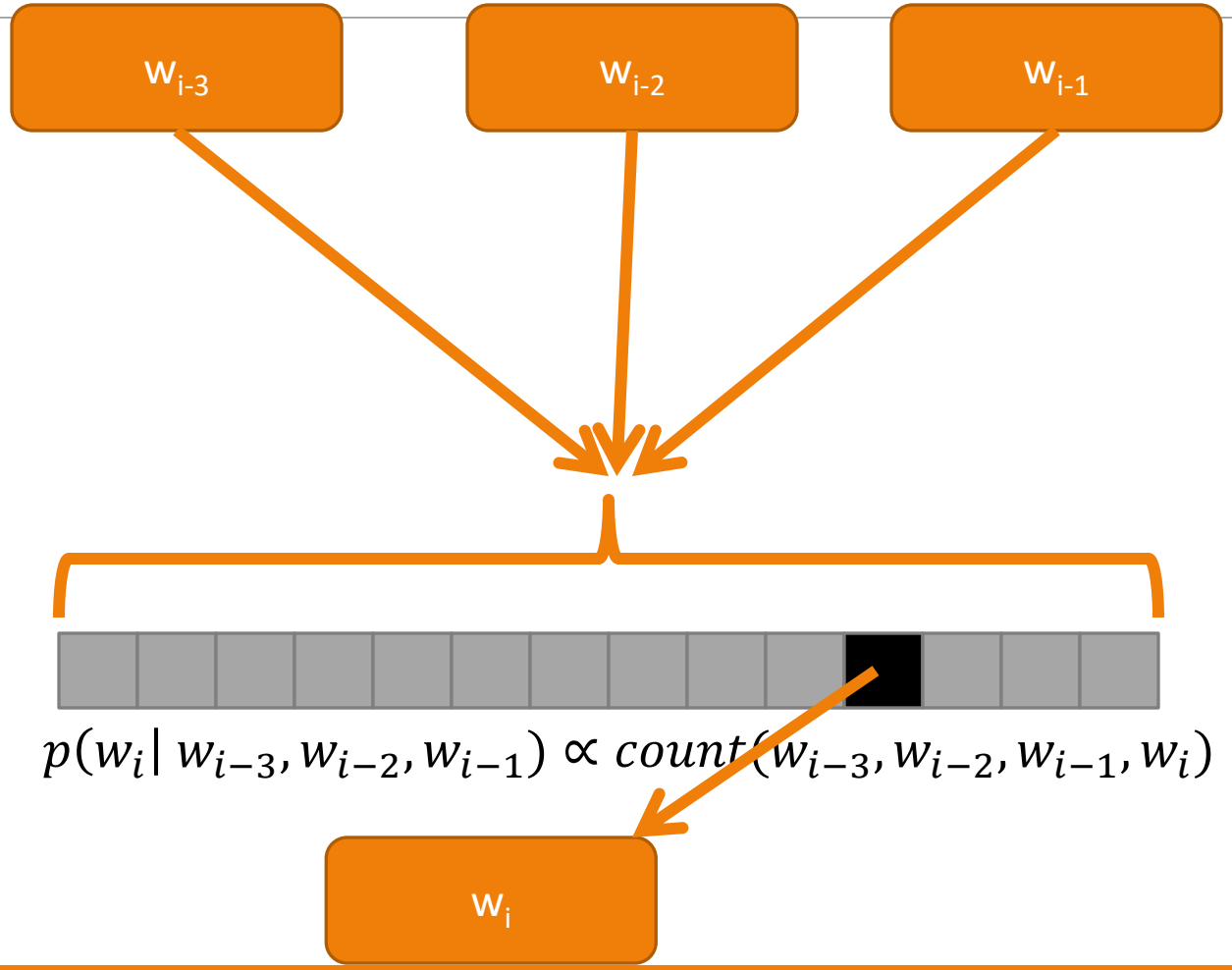
$$p(w_i \mid w_{i-3}, w_{i-2}, w_{i-1}) \propto count(w_{i-3}, w_{i-2}, w_{i-1}, w_i)$$

predict the next word

w_i

N-gram Language Models

given some context…



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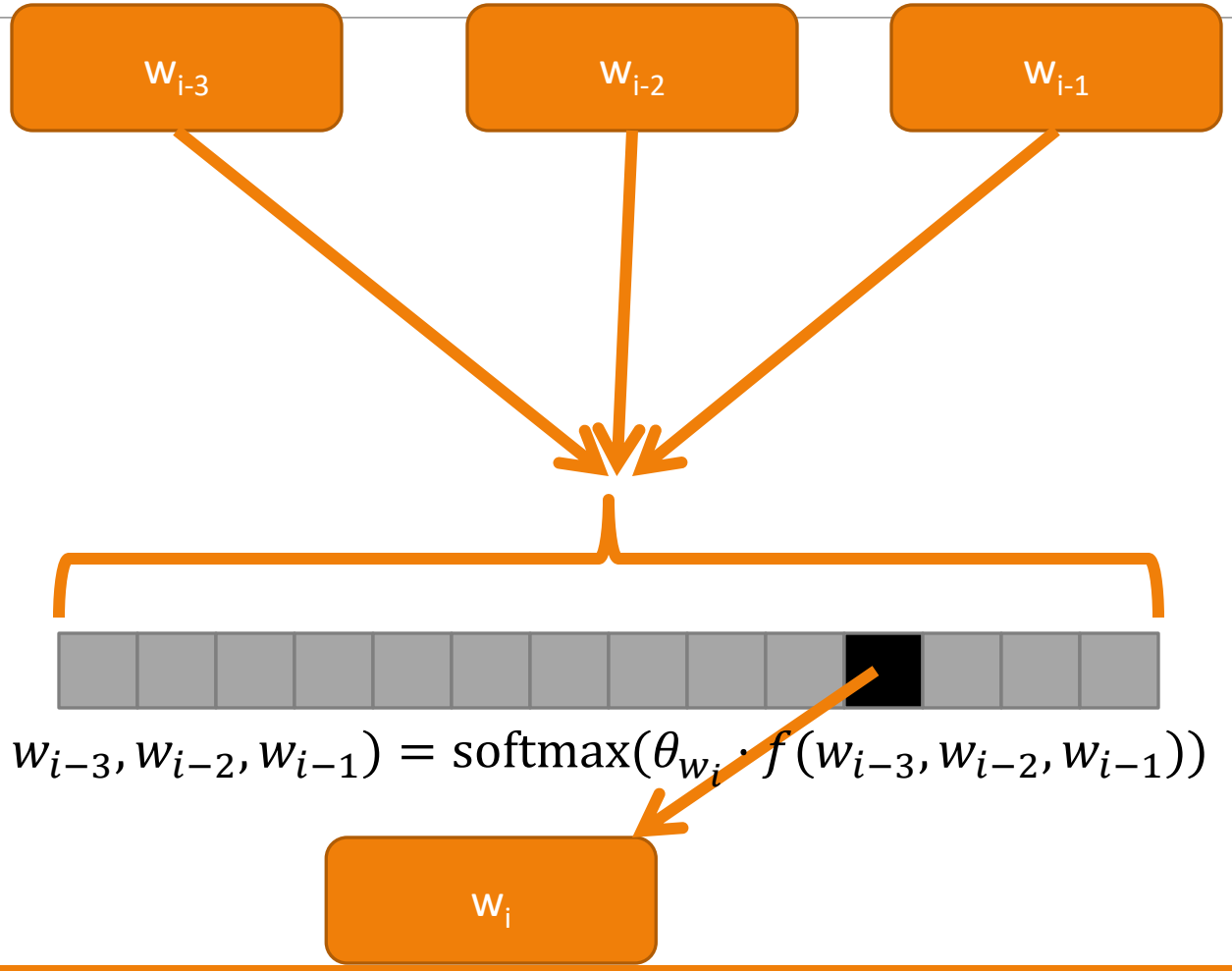
$$p(w_i \mid w_{i-3}, w_{i-2}, w_{i-1}) \propto count(w_{i-3}, w_{i-2}, w_{i-1}, w_i)$$

predict the next word

Review: Maxent Language Models

given some context…



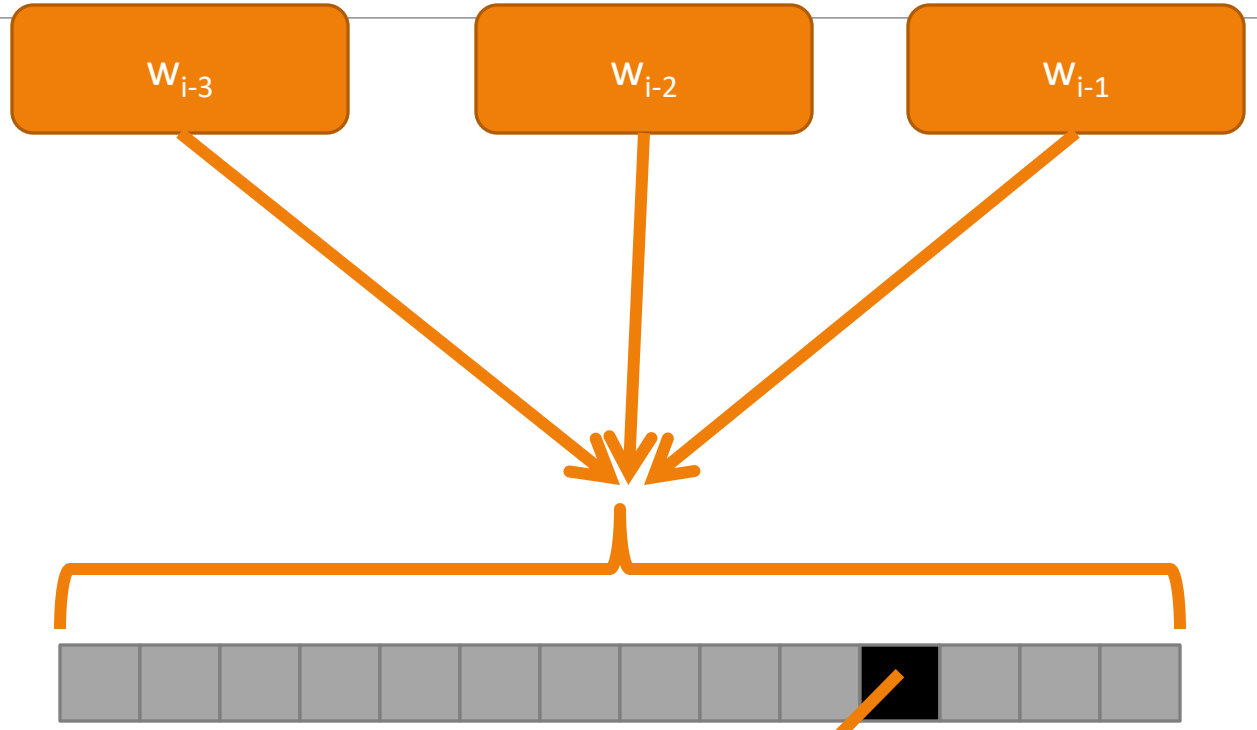
compute beliefs about what is likely…

$$p(w_i \mid w_{i-3}, w_{i-2}, w_{i-1}) = \text{softmax}(\theta_{w_i} \cdot f(w_{i-3}, w_{i-2}, w_{i-1}))$$

predict the next word

Maxent Language Models

given some context…



compute beliefs about what is likely…

$$p(w_i \mid w_{i-3}, w_{i-2}, w_{i-1}) = \text{softmax}(\theta_{w_i} \cdot f(w_{i-3}, w_{i-2}, w_{i-1}))$$

predict the next word

can we learn word-specific weights (by type)?

w_i

Neural Language Models

given some context…

w_{i-3} w_{i-2} w_{i-1}

can we *learn* the feature function(s) for *just* the context?

compute beliefs about what is likely…

$$p(w_i | w_{i-3}, w_{i-2}, w_{i-1}) = \text{softmax}(\theta_{w_i} \cdot f(w_{i-3}, w_{i-2}, w_{i-1}))$$

predict the next word

can we learn word-specific weights (by type)?

w_i

Types of Early LMs

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Featureful LMs

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Feedforward LMs

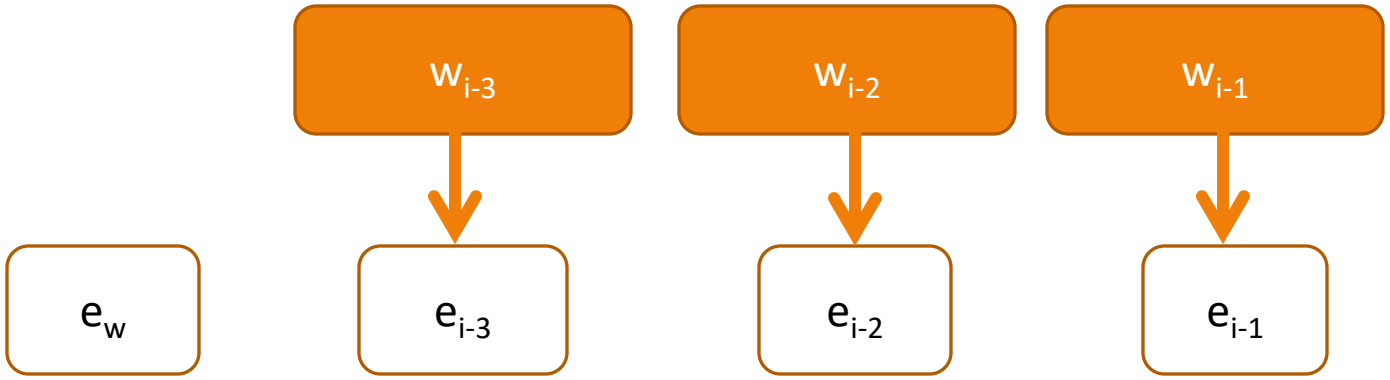
Recurrent/autoregressive NNs

Precursor to modern LMs

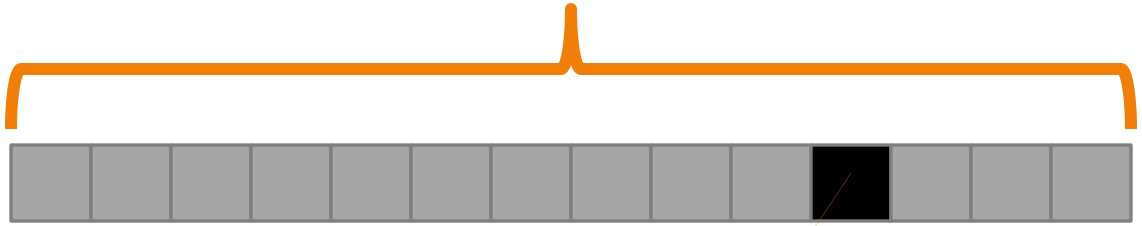
Neural Language Models

given some context…

create/use "distributed representations"…



compute beliefs about what is likely…



$$p(w_i|\, w_{i-3}, w_{i-2}, w_{i-1}) = \text{softmax}(\theta_{w_i} \cdot f(w_{i-3}, w_{i-2}, w_{i-1}))$$

predict the next word

w_i

Neural Language Models

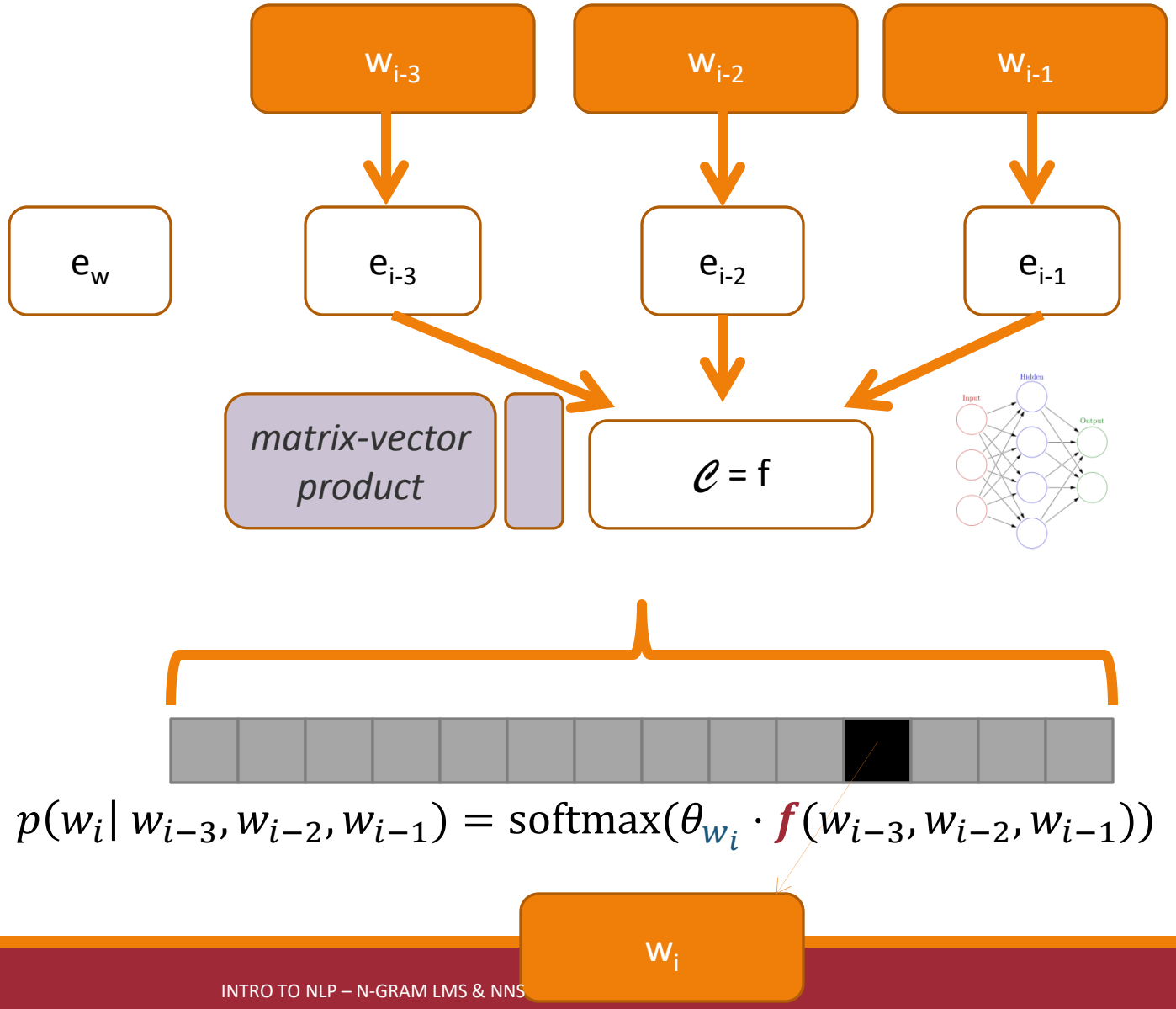
given some context…

create/use "distributed representations"…

combine these representations…

compute beliefs about what is likely…

predict the next word



$$p(w_i|\, w_{i-3}, w_{i-2}, w_{i-1}) = \text{softmax}(\theta_{w_i} \cdot f(w_{i-3}, w_{i-2}, w_{i-1}))$$

Neural Language Models

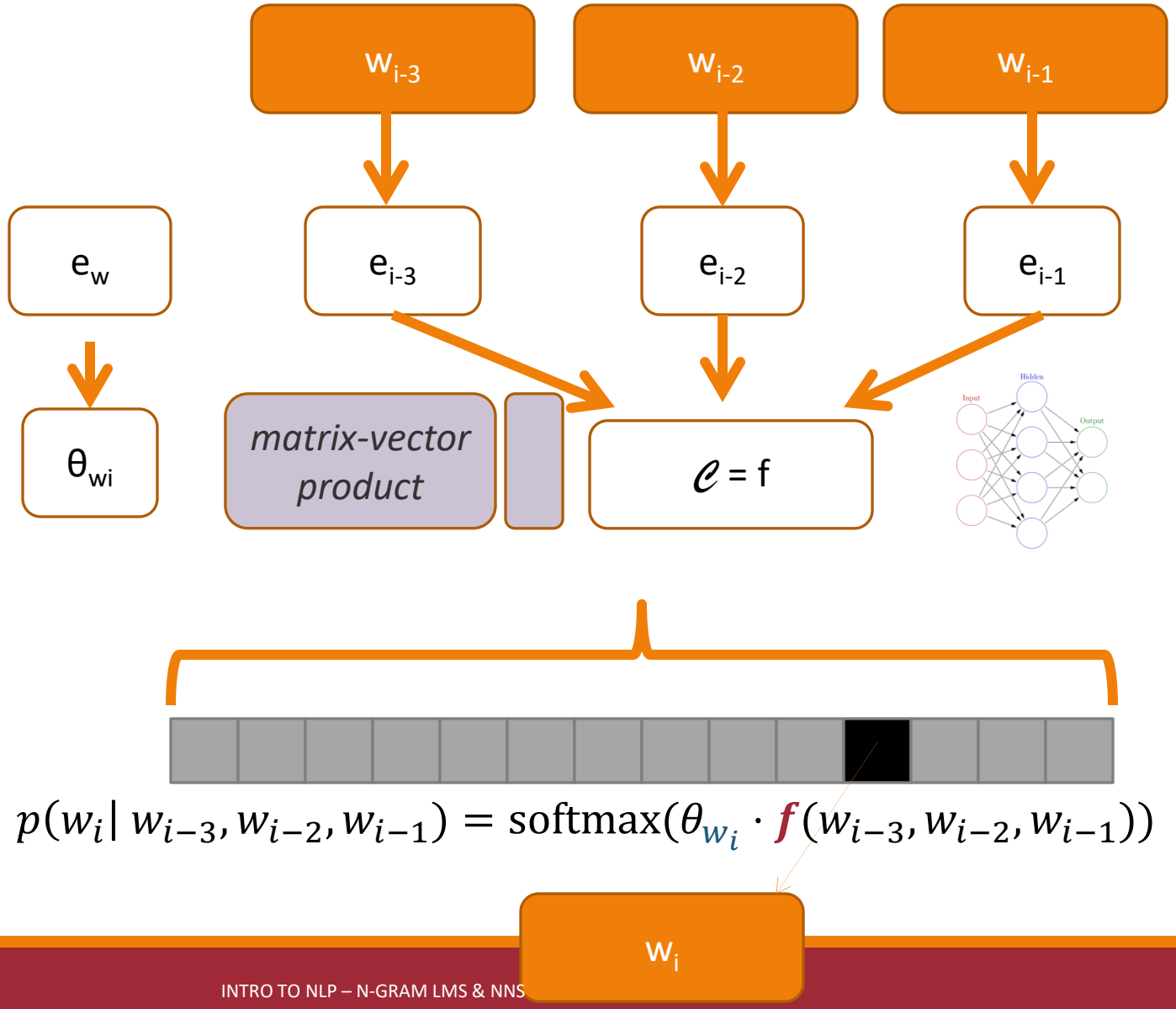
given some context…

create/use "distributed representations"…

combine these representations…

compute beliefs about what is likely…

predict the next word



$$p(w_i \mid w_{i-3}, w_{i-2}, w_{i-1}) = \text{softmax}(\theta_{w_i} \cdot f(w_{i-3}, w_{i-2}, w_{i-1}))$$

Neural Language Models

given some context…

create/use "distributed representations"…

combine these representations…

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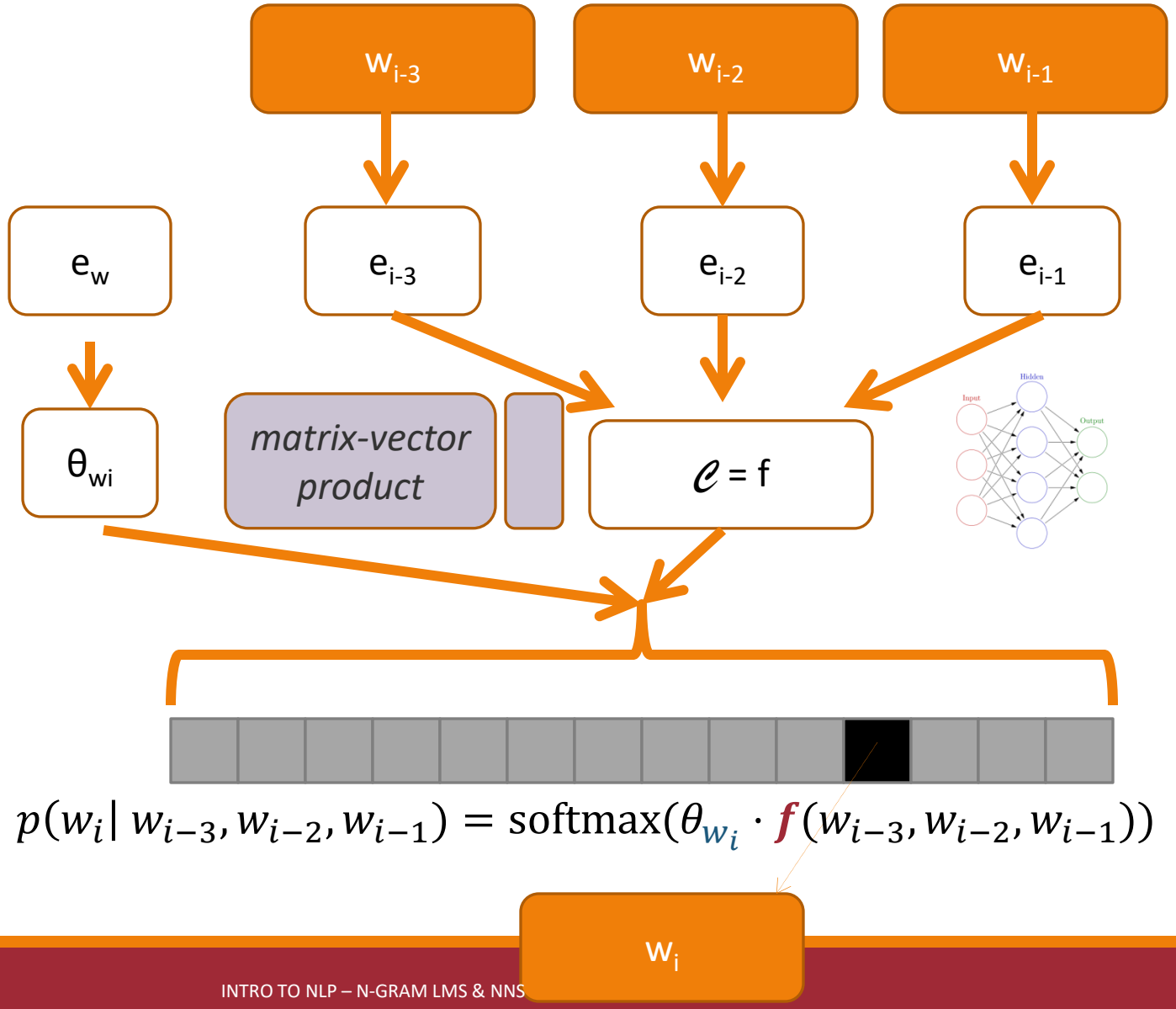
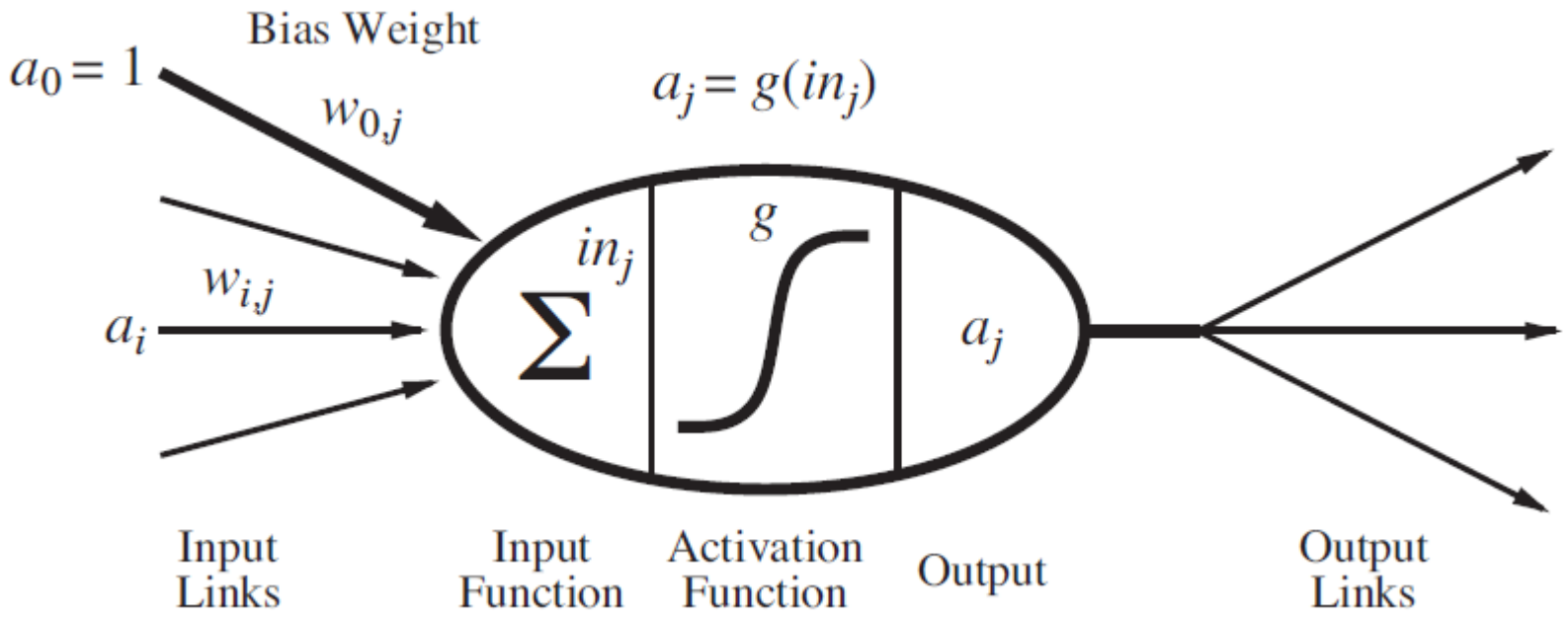
predict the next word



$$p(w_i \mid w_{i-3}, w_{i-2}, w_{i-1}) = \text{softmax}(\theta_{w_i} \cdot f(w_{i-3}, w_{i-2}, w_{i-1}))$$

Biologically-Inspired Learning Models: Neuron Unit

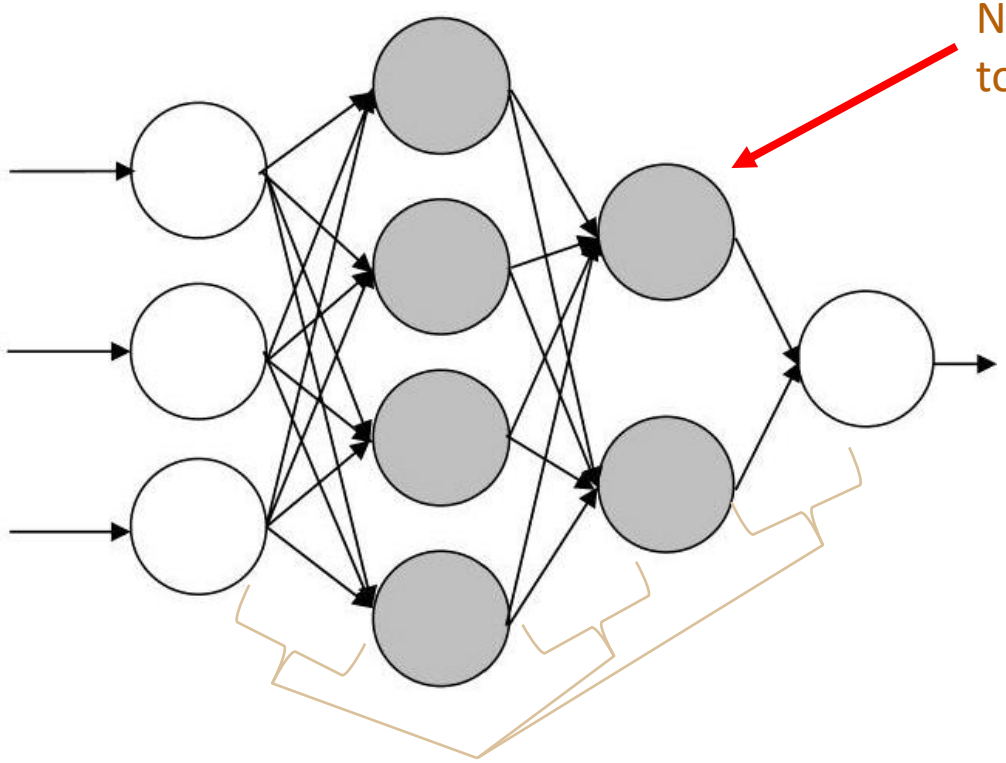
$$in_j = w_{0j} + w_{1j}a_1 + w_{2j}a_2 + \cdots + w_{ij}a_i$$



activations
$0 \leq a_i \leq 1$

weights
$-\infty < w_{ij} < \infty$

Multi-layer Networks: General Structure Example

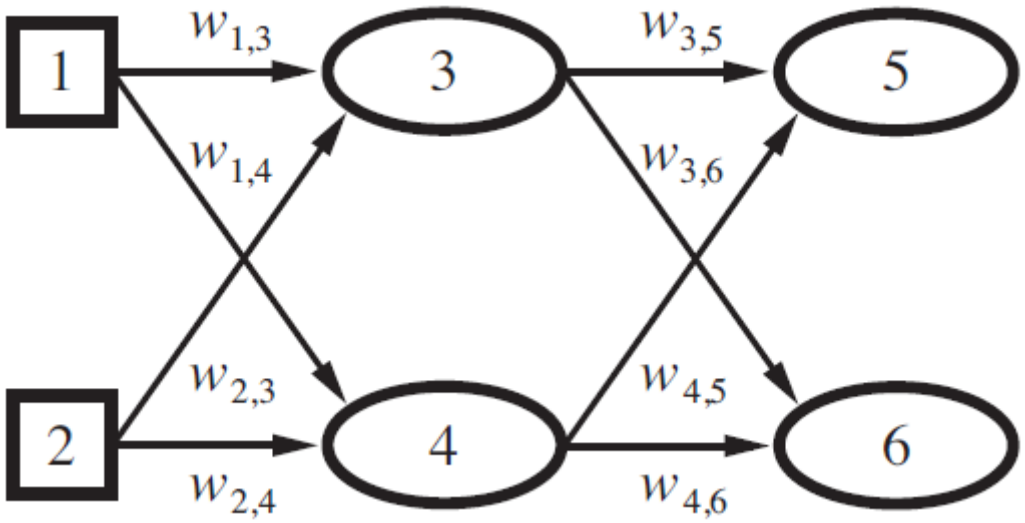


Note that the layers don't have to be the same size

Fully connected

Multi-layer Networks: General Structure

Multi-layer perceptrons (aka neural networks) will have **inputs**, one or more **hidden layers**, and an **output layer:**



Multi-layer Networks: General Structure

Multi-layer perceptrons (aka neural networks) will have **inputs**, one or more **hidden layers**, and an **output layer:**

Number of inputs, outputs, and number and size of hidden layers can vary

Combination of **different weights** and **different structures** represent different **functions**

We will treat each layer as **fully-connected**

- Each unit in one layer connects to every unit in the next layer

Computing Values: Forward Propagation

Forward propagation calculates the output values for a given set of input values

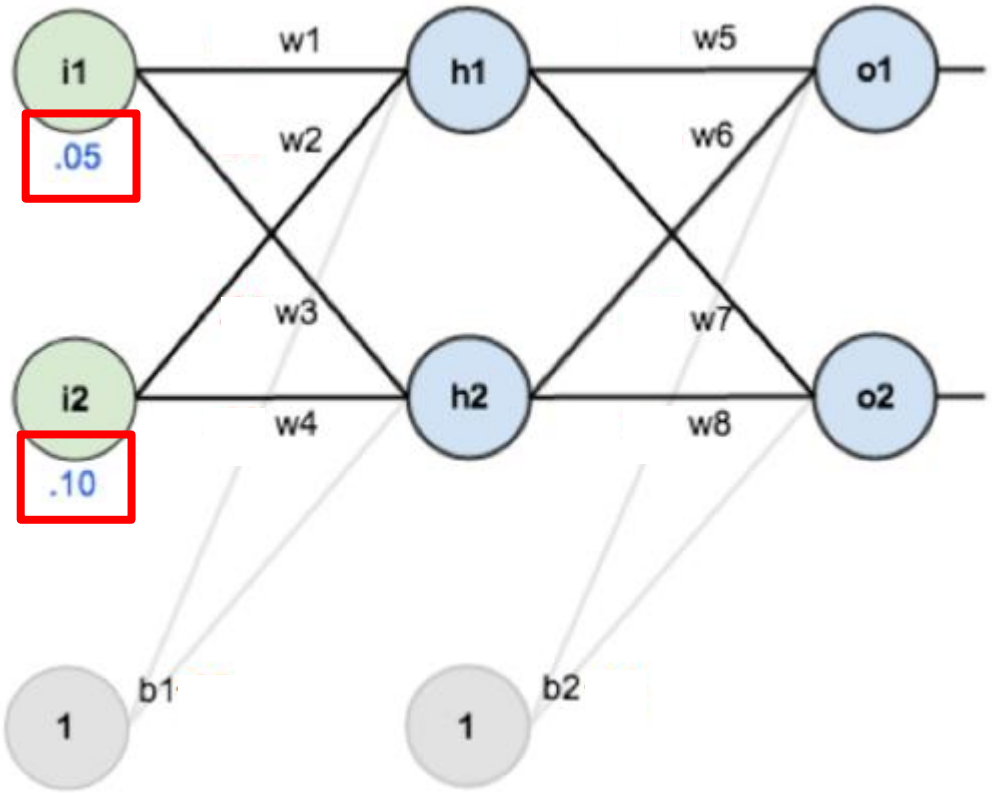
<u>Algorithm</u>

For each layer:

1. Calculate the weighted sum of inputs to each neuron unit
2. Evaluate the activation function to determine the output of each neuron unit
3. Use outputs as inputs for the next layer

Forward Propagation Example

Calculate the output of the network below, assuming each neuron uses a sigmoid activation function, given 0.05 and 0.1 as inputs.

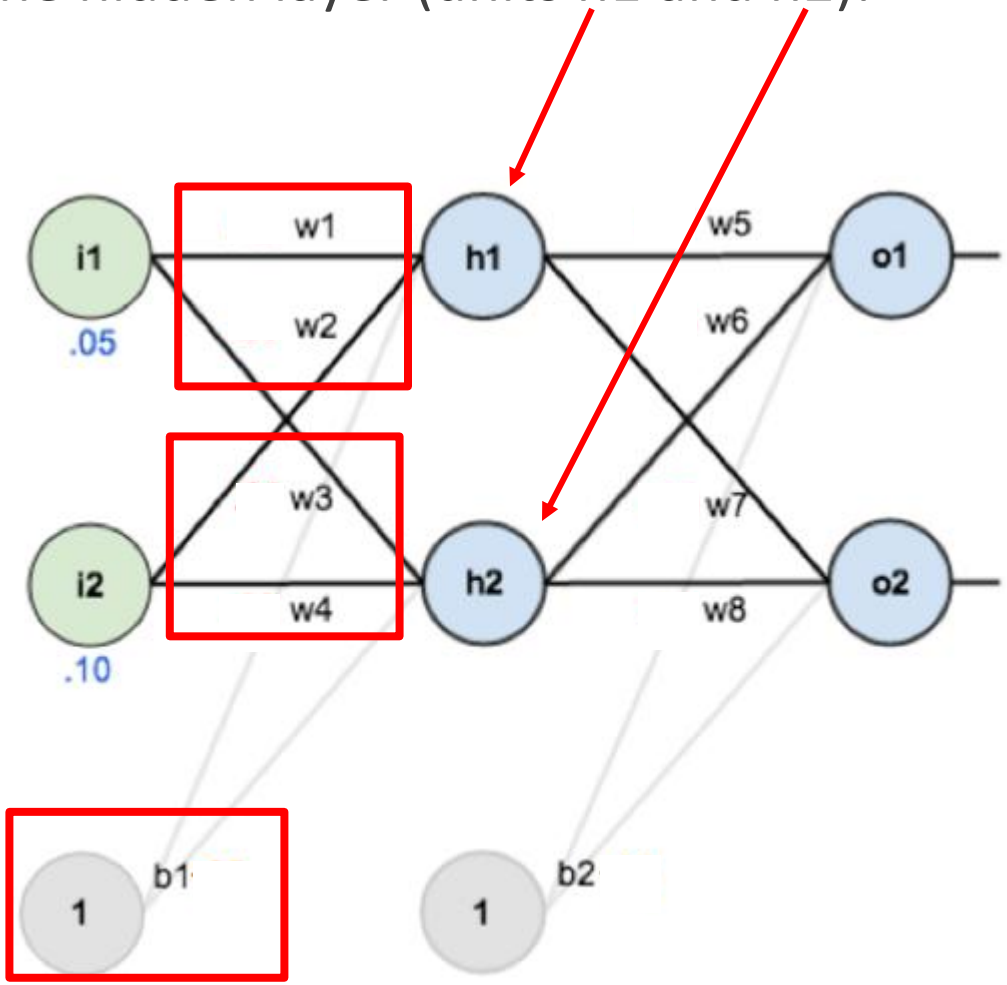


For each layer:

1. Calculate the weighted sum of inputs to each neuron unit
2. Evaluate the activation function to determine the output of each neuron unit
3. Use outputs as inputs for the next layer

Forward Propagation Example

Calculate inputs to the hidden layer (units h1 and h2):



$$in_{h1} = w_1 i_1 + w_2 i_2 + b_1$$
$$= .15(.05)+.2(.1)–.35$$
$$= .0075+.02–.35$$
$$= –.3225$$

$$in_{h2} = w_3 i_1 + w_4 i_2 + b_2$$
$$= .25(.05)+.3(.1)–.35$$
$$= .0125+.03–.35$$
$$= –.3075$$

For each layer:

1. Calculate the weighted sum of inputs to each neuron unit
2. Evaluate the activation function to determine the output of each neuron unit
3. Use outputs as inputs for the next layer

Forward Propagation Example

Calculate <u>outputs</u> to the hidden layer (units h1 and h2):

How do we do this?
Use our activation function!

$$g(x) = \frac{1}{1 + e^{-x}}$$

What will be our *x*?

in_{h1} =–.3225
in_{h2} = –.3075

For each layer:
1. Calculate the weighted sum of inputs to each neuron unit
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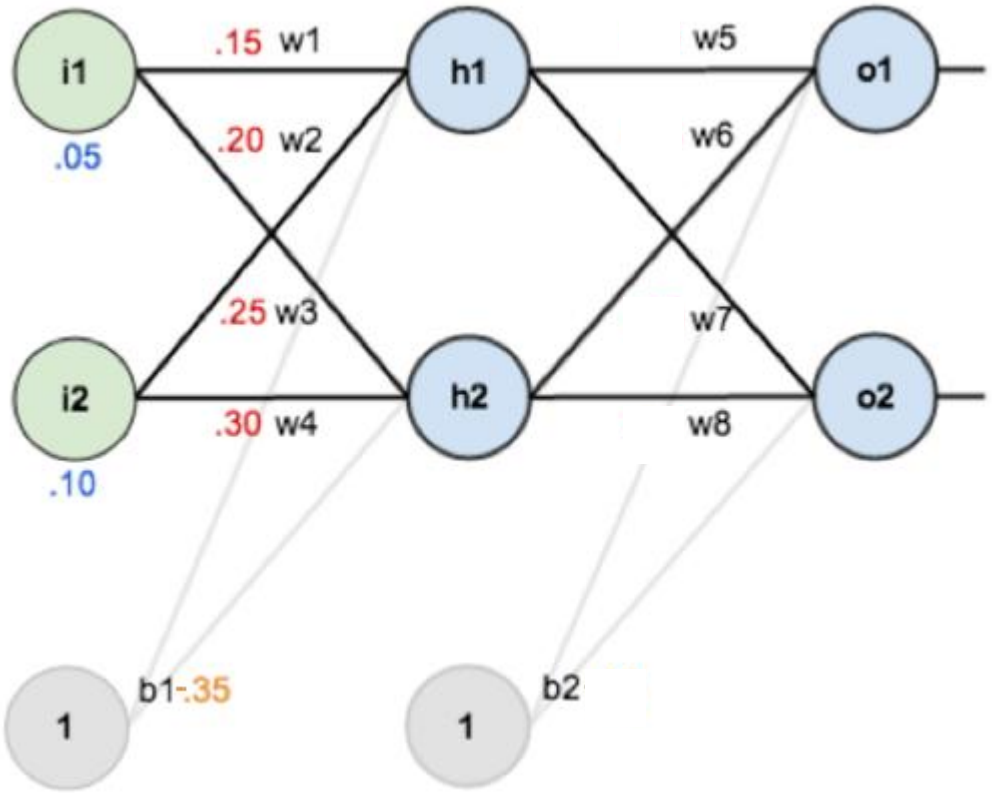
3. Use outputs as inputs for the next layer



$out_{h1} = g(in_{h1})$
$= \frac{1}{1+e^{-in_{h1}}}$
$= \frac{1}{1+e^{-(-.3275)}}$
$= .4188$

$out_{h2} = g(in_{h2})$
$= \frac{1}{1+e^{-in_{h2}}}$
$= \frac{1}{1+e^{-(-.3075)}}$
$= .4237$

How are Neural Networks used?

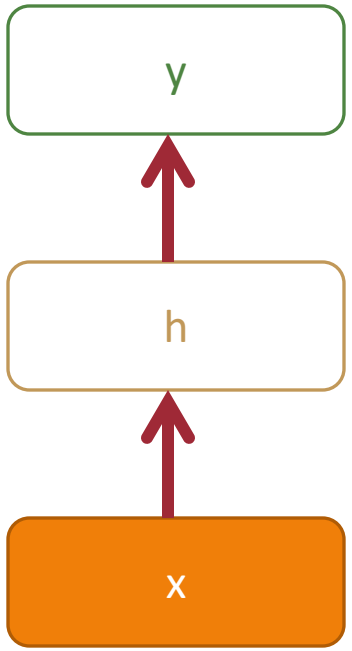
Are neural networks supervised or unsupervised learning?

- Inputs to the network are features of our data set
- Outputs to the network are our labels

Can they be used for classification or regression?

- Either!

Network Types: Flat Input, Flat Output



1. Feed forward

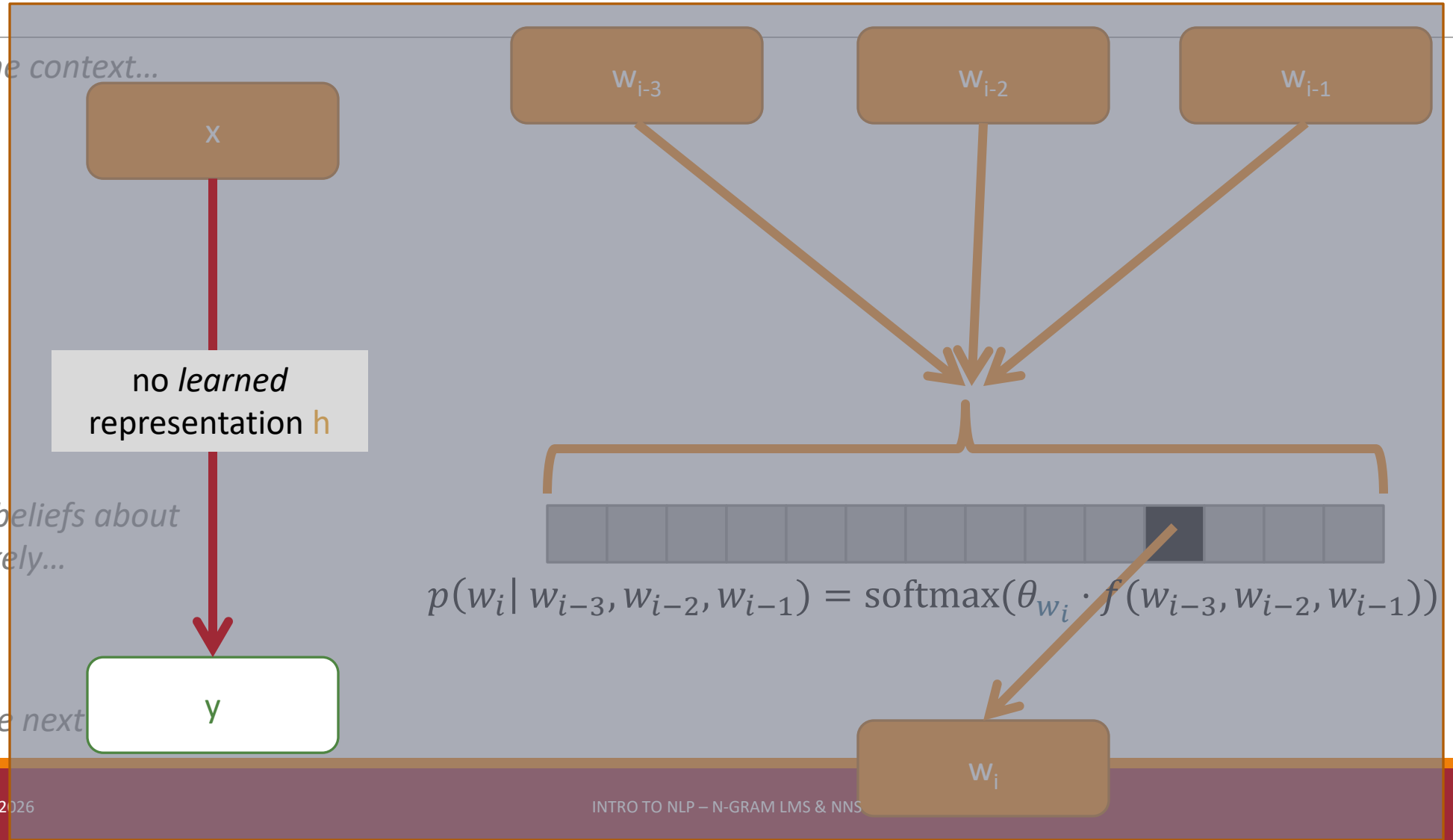
Linearizable feature input
Bag-of-items classification/regression
Basic non-linear model

Maxent Language Models

given some context…

compute beliefs about what is likely…

predict the next



$$p(w_i \mid w_{i-3}, w_{i-2}, w_{i-1}) = \text{softmax}(\theta_{w_i} \cdot f(w_{i-3}, w_{i-2}, w_{i-1}))$$

Neural Language Models

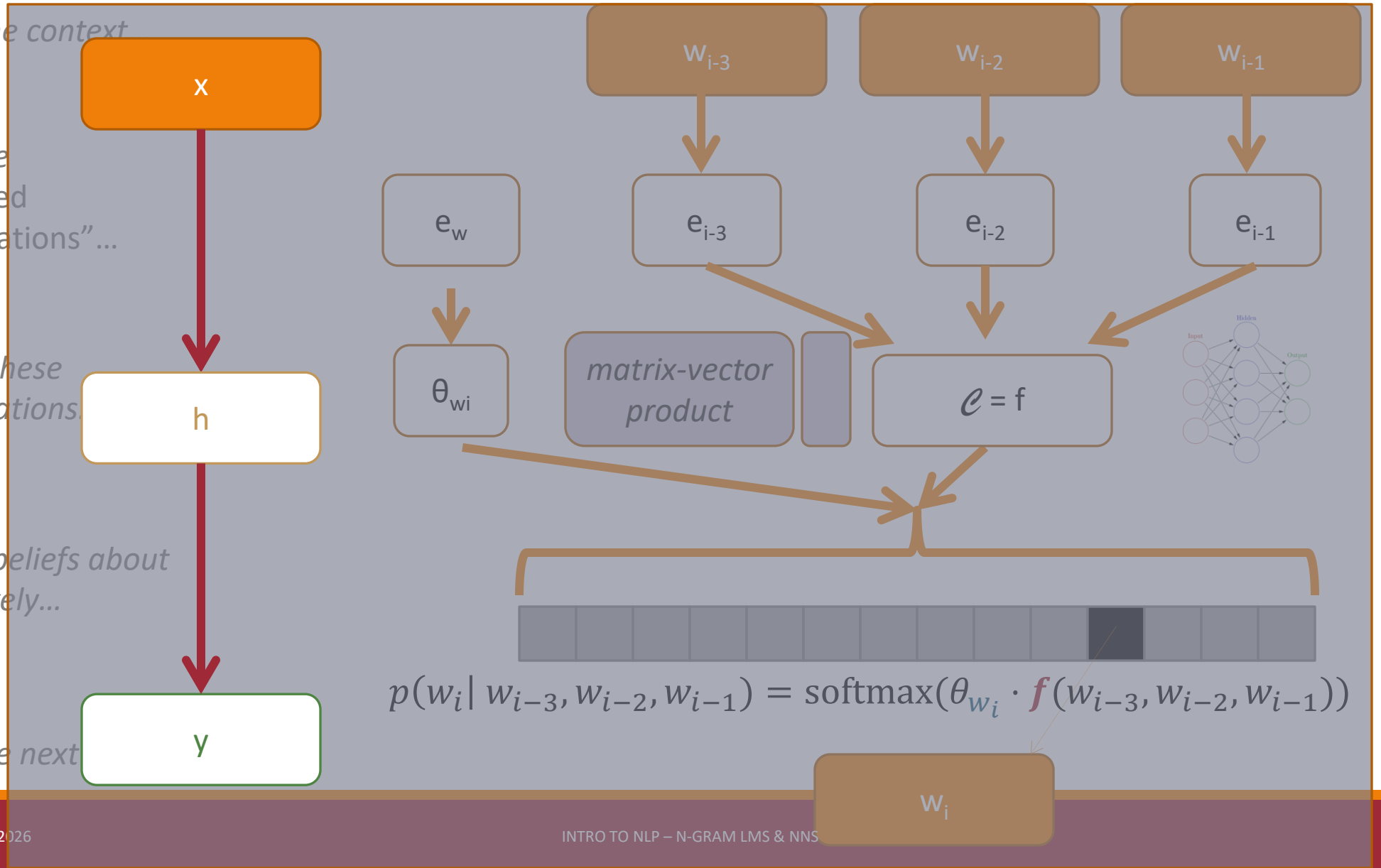
given some context

create/use "distributed representations"…

combine these representations

compute beliefs about what is likely…

predict the next



Common Types of Flat Input, Flat Output

Feed forward networks

Multilayer perceptrons (MLPs)

General Formulation:

Input: x
Compute:

$h_0 = x$
for layer l = 1 to L:
$h_l = f_l(W_l\, h_{l-1} + b_l)$ linear layer

hidden state at layer l

(non-linear) activation function at l

return $\underset{y}{\operatorname{argmax}}\, \operatorname{softmax}(\theta h_L)$

In Pytorch (torch.nn):

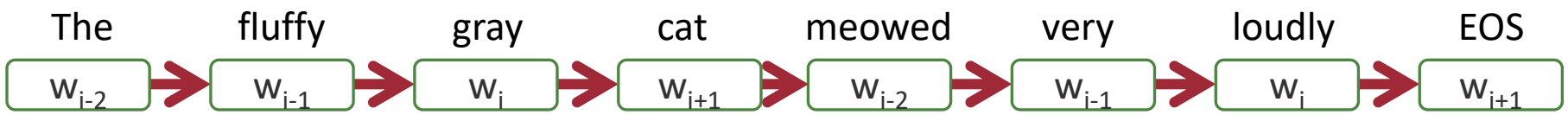
Activation functions:
https://pytorch.org/docs/stable/nn.html?highlight=activation#non-linear-activations-weighted-sum-nonlinearity

Linear layer:
https://pytorch.org/docs/stable/nn.html#linear-layers

torch.nn.Linear(
in_features=<dim of h_{l-1} >,
out_features=<dim of h_l >,
bias=<Boolean: include bias b_l >)

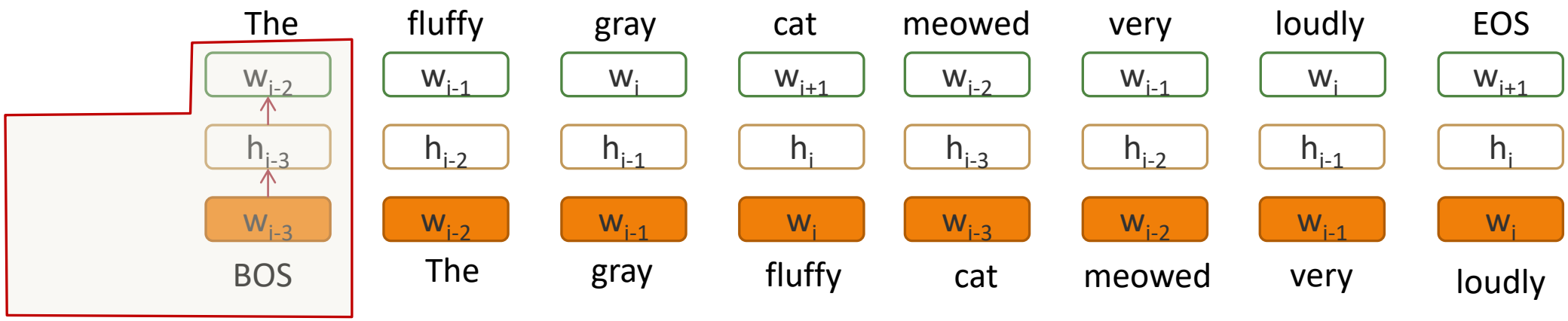
A Neural N-Gram Model

The fluffy gray cat meowed very loudly



A Neural N-Gram Model (N=3)

The fluffy gray cat meowed very loudly



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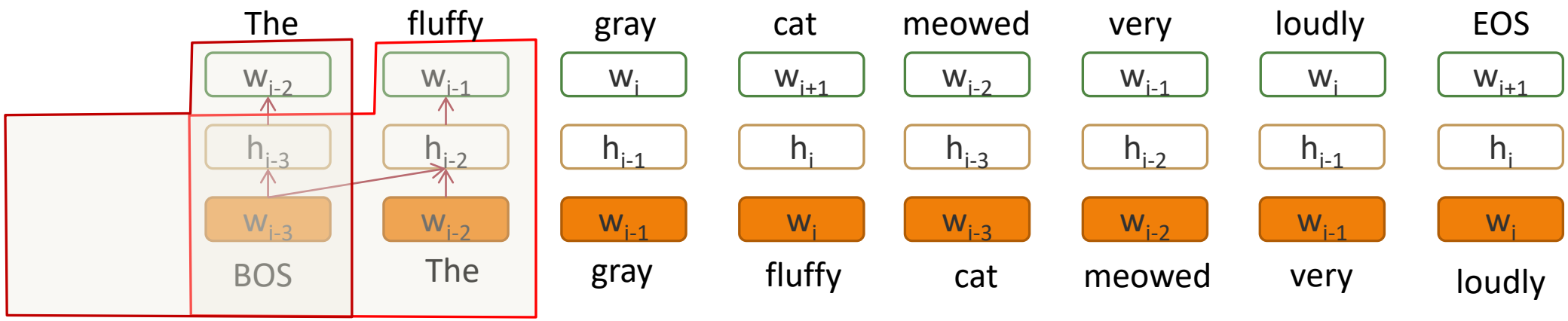

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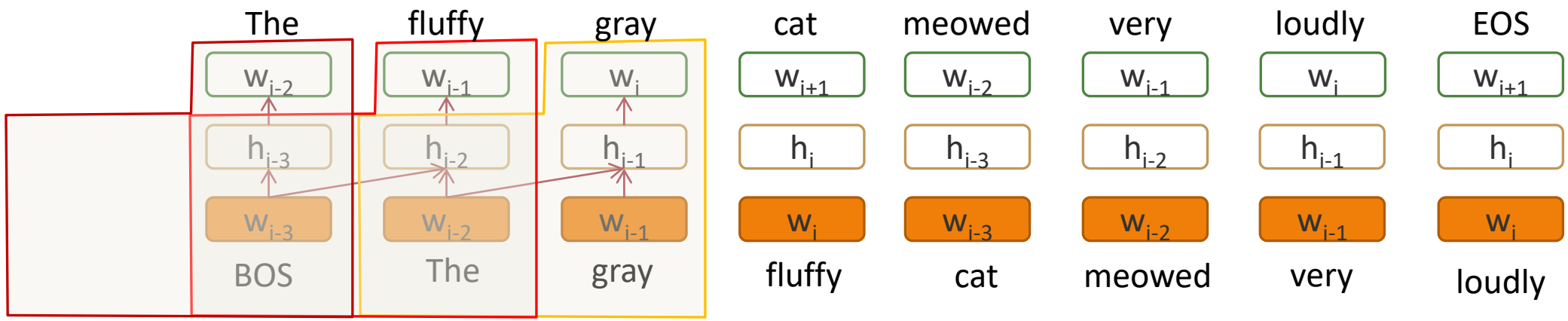

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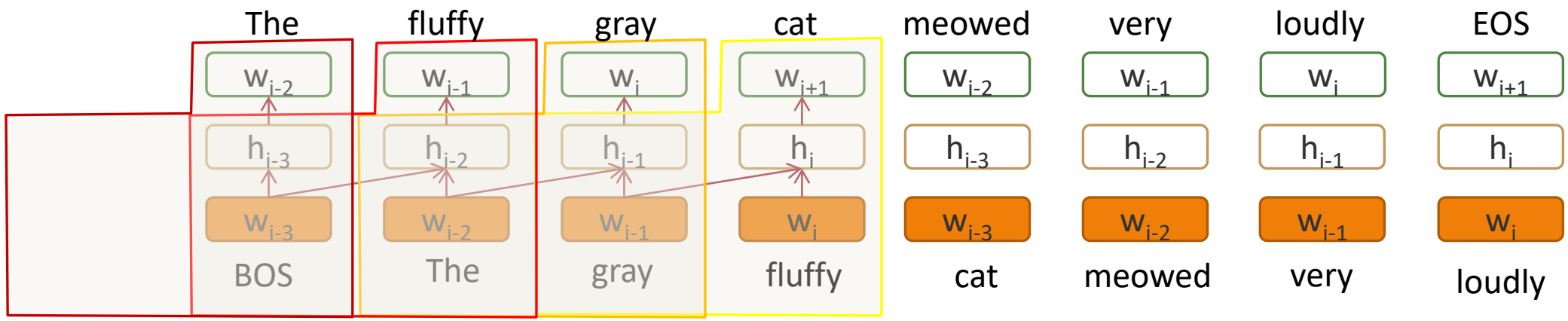

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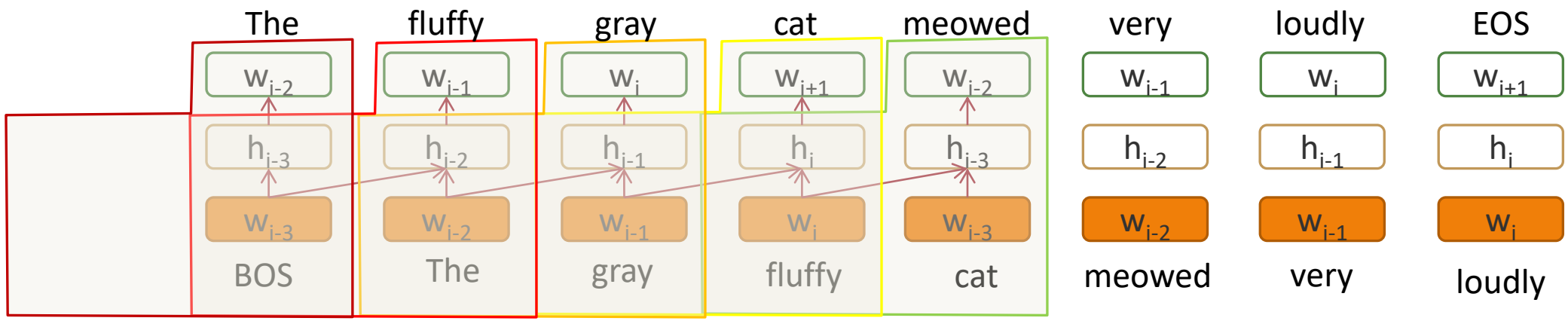

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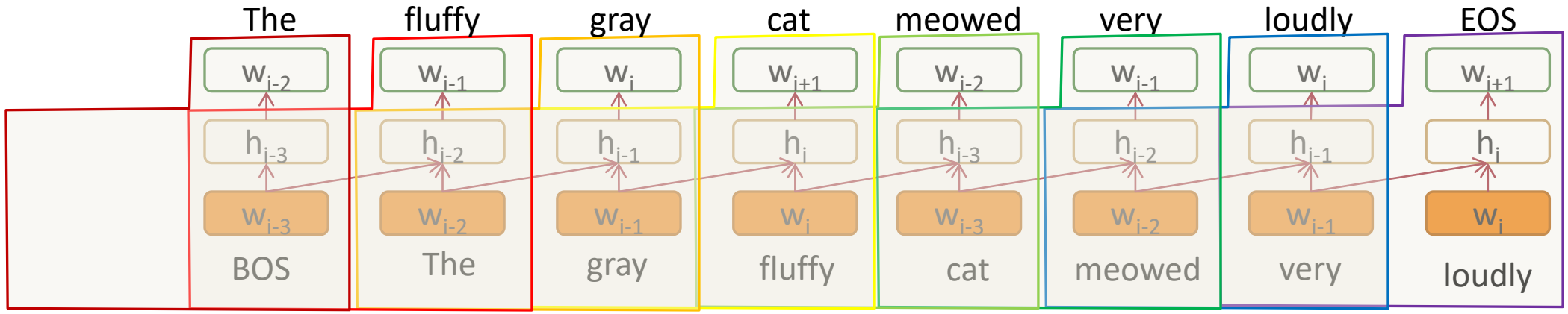
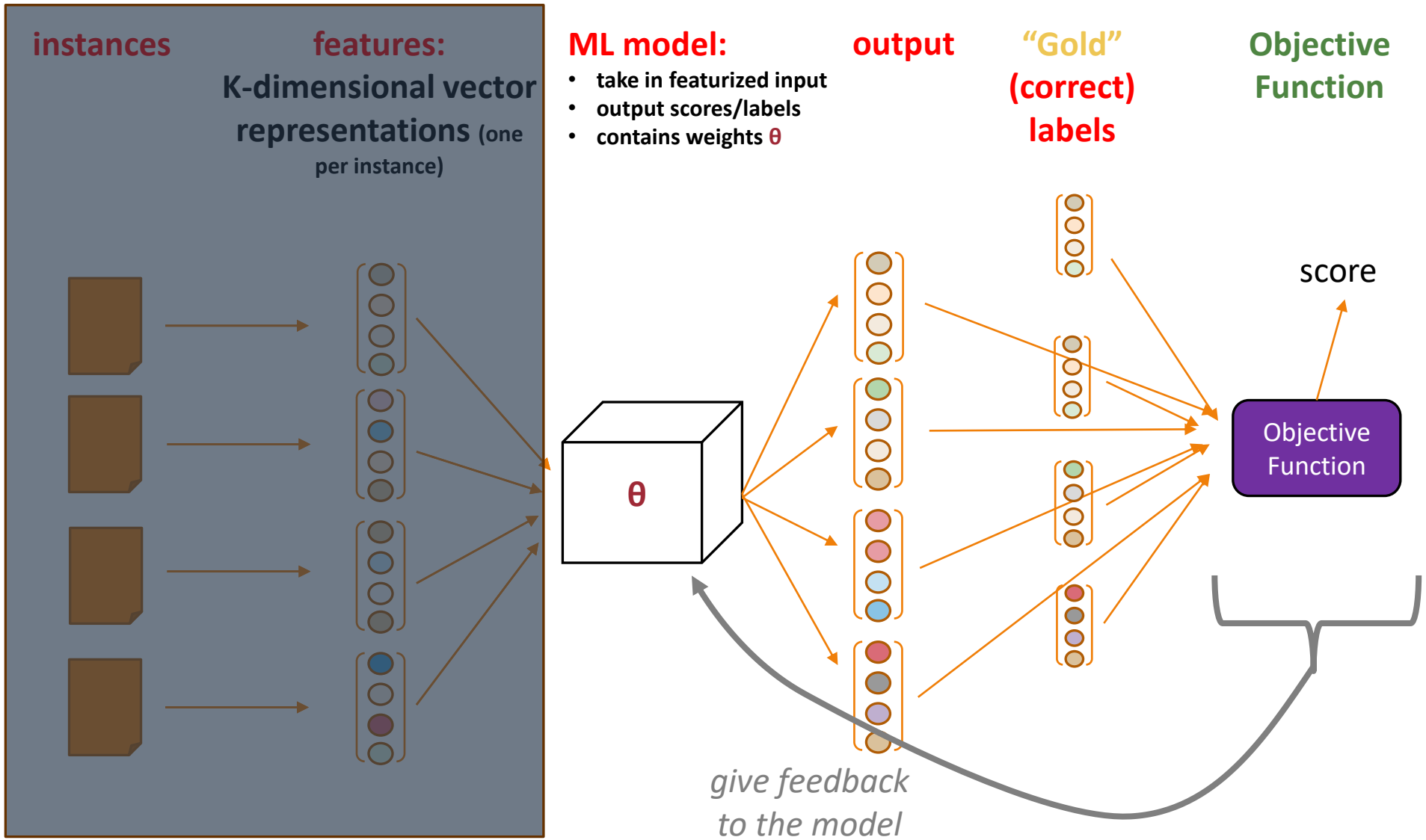


ML/NLP Framework for Learning



Review:
Maximize Log-Likelihood (Classification)

Original maxent equation

$$\frac{\exp(\theta_y^T f(x))}{\sum_{y'} \exp(\theta_{y'}^T f(x))}$$

$$\log \prod_i p_\theta(y_i|x_i) = \sum_i \log p_\theta(y_i|x_i)$$

Differentiating this becomes nicer (even though Z depends on θ)

Inverse of exp
log(exp(x)) = x

$$= \sum_i \overbrace{\theta_{y_i}^T f(x_i) - \log Z(x_i)}$$

$$= F(\theta)$$

Review:
Minimize Cross Entropy Loss

Classifier output

True probability (i.e., correct output)

$$L^{\text{xent}}(\hat{y}, y) =$$

index of "1" indicates correct value

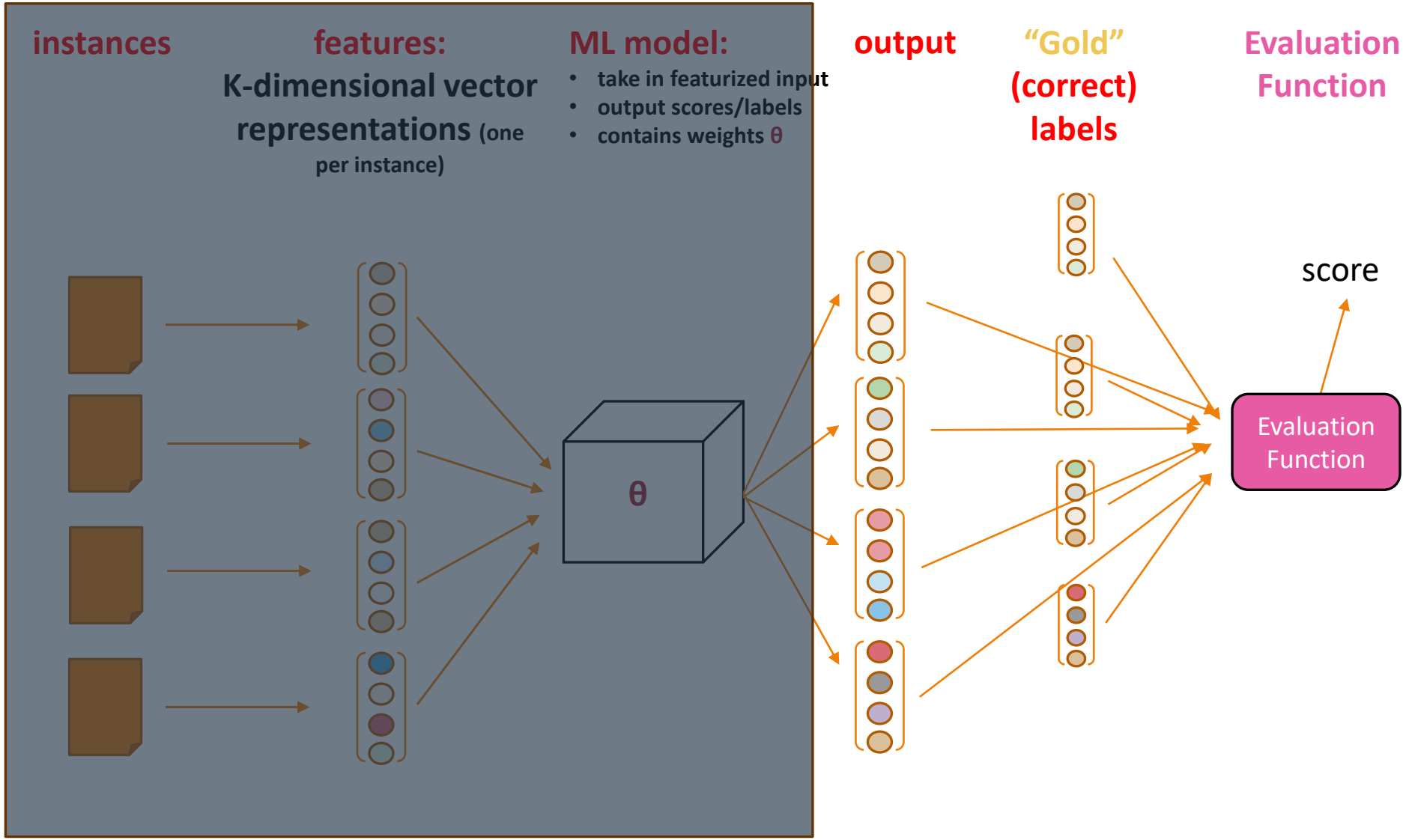
$$\begin{pmatrix} 0 \\ 0 \\ \dots \\ 1 \\ \dots \\ 0 \end{pmatrix}$$

one-hot vector

Cross entropy:
How much $\hat{y}$ differs from the true y

objective is convex
(when f(x) is not learned)

ML/NLP Framework for Prediction



What are some examples of evaluation functions?

“A Neural Probabilistic Language Model”

Bengio et al. (2003)

BASELINES

LM Name	N-gram	Params.	Test PPL
Interpolation	3	---	336
Kneser-Ney backoff	3	---	323
Kneser-Ney backoff	5	---	321
Class-based backoff	3	500 classes	312
Class-based backoff	5	500 classes	312

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NPLM

N-gram	Word Vector Dim.	Hidden Dim.	Mix with non-neural LM	PPL
5	60	50	No	268
5	60	50	Yes	257
5	30	100	No	276
5	30	100	Yes	252

“we were not able to see signs of over- fitting (on the validation set), possibly because we ran only 5 epochs (over 3 weeks using 40 CPUs)” (Sect. 4.2)

A Closer Look at Neural $p($ Won't you please donate? $|$ Primary $)$

This is a *class-based* language model, but incorporate the label into the *embedding representation*

To learn $p($ Won't you please donate? $|$ Class $)$:

Define an embedding method that makes use of the specific label Class

Unlike count-based models, you don't *need* "separate" models here

LM Comparison for $p($ Won't you please donate? $|$ Primary $)$

N-GRAM/COUNT-BASED	MAXENT/LR	NEURAL
Class-specific	Class-based	Class-based
	Uses features	Uses *embedded* features